

# Audit Descent Calculus

Certificate Naturality, Observation Galois Lattices, Atomic Rigidity, and an  
Executable Scientific Court

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## Contents

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1. Executive resolution and status ledger
2. Scope and scientific court
3. Audit descent datum
4. Observation-descent Galois lattice
5. Quantitative defect transport
6. Arithmetic atomic rigidity
7. Consequences for the arithmetic program
8. Catalogue-wide mathematical corrections
9. Executable audit engine
10. Novelty audit
11. Failure modes and falsifiers
12. Publication readiness and highest-leverage next step
13. Open problems, references, and appendices

## Executive resolution

The framework contains a viable mathematical core, but its next foundation is not another global physical analogy or spectral ansatz. It is a calculus of **certificate transport**: a way to decide whether a verified claim remains verified when one changes scale, instrument, quotient, boundary completion, prime set, or cutoff.

This paper defines that calculus. At each context it places a finite exact chain complex of claims, obligations, and certificates. A context change is a graded map. Its failure to commute with verification is the **certificate-interchange defect**

$$\Theta_{\alpha,n} = \partial_n^d T_{\alpha,n} - T_{\alpha,n-1} \partial_n^c.$$

Vanishing of all such defects is exactly the condition required for verified classes to descend to homology after transport. A nonzero column is an exact, finite witness identifying an obligation that fails to survive the context change. For two paths between the same contexts, the difference of their composite transports is a path-holonomy certificate. The previously isolated prime-adjunction cocycle is one specialization.

The second foundation is an observation-descent lattice. Experimental indistinguishability is encoded by a relation, and a query is legal on the resulting quotient exactly when it is constant on that relation. This gives an antitone Galois connection between observation relations and admissible queries, with explicit counterexample pairs when descent fails.

The arithmetic trace program exposes a third, previously unsealed boundary. Smooth counterterms that are admissible at every finite cutoff can concentrate into off-origin Dirac masses in the limit and thereby reconstruct the forbidden prime-power comb. This paper introduces an off-origin atomic charge and a computable nonconcentration modulus. The resulting atomic-rigidity theorem forces prime-power atoms to arise from the raw distributional trace rather than being smuggled through the counterterm channel.

These structures have been implemented as version 0.3.0a2 of a standard-library, exact-rational audit engine. The implementation checks claim manifests, evidence-derived product-valued hard gates, certificate complexes, context transport, path holonomy, observation descent, atomic-concentration records, arithmetic no-go conditions, schema/runtime parity, null discrimination, and exact propagation of quantitative defect bounds. More than sixty deterministic tests pass. This establishes an executable structural kernel. It does not prove the Riemann hypothesis, construct the required infinite-dimensional operator, or validate every scientific claim in the source catalogue.

## Resolution ledger

| Object                                                           | Status                                    | Resolution                                                             |
|------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------|
| Certificate-interchange identity and induced homology transport  | Proven                                    | Exact finite-dimensional algebra                                       |
| Composition law for certificate defects                          | Proven                                    | Exact derivation identity                                              |
| Observation relation/query Galois connection                     | Proven                                    | Elementary order-theoretic theorem                                     |
| Affine transport-defect composition                              | Proven                                    | Associative certificate monoid                                         |
| Off-origin atomic-charge invariance                              | Proven on the declared distribution class | Excludes admissible counterterms and finite analytic traces            |
| Atomic-rigidity criterion                                        | Proven                                    | Requires local variation control and a vanishing concentration modulus |
| Catalogue-wide audit calculus as a useful unifying foundation    | Strongly supported                        | Implemented on finite exact records; universality is not claimed       |
| Effective rational spline-mollifier dictionary                   | Plausible; unresolved                     | The density theorem needs an explicit quasi-interpolant and constants  |
| Certified interval Weil Gram engine                              | Plausible; unresolved                     | Specification is mature; complete implementation remains               |
| Infinite-dimensional prime-local operator fold                   | Outside current knowledge                 | Self-adjoint construction and joint trace-class limit remain open      |
| Optical analogue implies an arithmetic trace construction        | Refuted as an implication                 | It is an architectural analogy and conservation benchmark only         |
| Simulation-grid or scale-torsion inference from spectral leakage | Ill-posed without a null model            | No admissible estimator or identifiability theorem has been supplied   |

## 1. Scope and scientific court

### 1.1 Typed claim states

A claim has three independent coordinates.

1. **Mathematical type:** definition, theorem, theorem schema, conjecture, diagnostic, empirical claim, analogy, or open problem.
2. **Epistemic state:** declared, checked, empirically passed, or externally replicated.
3. **Deployment state:** research-only, sandboxed, candidate, admitted, or retired.

Each prospective gate has one of four states:

$$G = \{ \perp, Pass, Fail, Conflict \}.$$

Here  $\perp$  means unrun. A promoted claim carries a product  $g \in G^I$ , one coordinate per preregistered gate. Fatal gates are conjunctive: admission requires every fatal coordinate to equal *Pass*. A conflict is preserved as a conflict; inconsistent enclosures or certificates are never averaged.

This product replaces a single maturity ladder. A theorem can be mathematically checked but operationally sandboxed. An experiment can be externally replicated while a proposed causal interpretation remains research-only. Soft scores may rank only the surviving fiber after fatal gates have passed.

### 1.2 Fatality belongs to claim dependencies

Fatal failure is not generally an ideal in an operator algebra. For example, two operators can separately violate a drift gate while their sum cancels the drift. Closure under addition therefore fails. Fatality is instead represented by an upward-closed filter in an acyclic dependency graph.

Let  $\Gamma = (V, E)$  be a directed acyclic graph whose edges  $v \rightarrow w$  mean that claim  $w$  depends on obligation  $v$ . For failed fatal gates  $F \subset V$ , define the affected set

$$Up_{\Gamma}(F) = \{ w \in V : \text{there is a directed path from some } f \in F \text{ to } w \}.$$

A root claim is demoted exactly when it lies in this set. This is a provenance rule, not a new physical algebra.

## 2. Audit descent datum

### 2.1 Contexts

A **context category**  $C$  has objects representing declared scientific contexts and arrows representing legal changes. Examples include:

- a scale refinement;
- replacement of an instrument by a calibrated successor;
- passage from raw state to an observation quotient;
- enlargement of a boundary model;
- adjoining a prime to a finite place set;
- increasing a numerical cutoff;
- translating a deterministic model to a stochastic observation kernel.

The category must be specified. Its arrows are not assumed invertible. Different arrows with the same endpoints may represent genuinely different procedures.

### 2.2 Certificate complexes

For each context  $c \in C$ , let

$$\cdots \rightarrow K_2(c) \xrightarrow{\partial_2^c} K_1(c) \xrightarrow{\partial_1^c} K_0(c)$$

be a finite chain complex over a declared computable exact field  $F$ , with

$$\partial_{n-1}^c \partial_n^c = 0.$$

The semantics of each degree must be fixed in the manifest. A useful convention is:

- high degrees: primitive data, calibration records, and atomic certificates;
- intermediate degrees: lemmas, compatibility obligations, and boundary terms;
- degree zero: promoted claims or output assertions.

The homology group

$$H_n(K(c)) = \ker \partial_n^c / \operatorname{im} \partial_{n+1}^c$$

records obligations closed under the declared verification rules but not discharged by lower-level certificates. This interpretation is relative to the manifest; homology is not automatically scientific truth.

## 2.3 Context transport

For an arrow  $\alpha : c \rightarrow d$ , a proposed certificate transport is a graded family

$$T_{\alpha,n} : K_n(c) \rightarrow K_n(d).$$

It need not initially be a chain map. That property is audited rather than assumed.

### Definition 2.1 — Certificate-interchange defect

The degree- $n$  interchange defect is

$$\Theta_{\alpha,n} = \partial_n^d T_{\alpha,n} - T_{\alpha,n-1} \partial_n^c.$$

It measures the noncommutation of “transport, then verify” and “verify, then transport.”

### Theorem 2.2 — Lawful promotion transport

The transport  $T_\alpha$  induces homomorphisms

$$H_n(T_\alpha) : H_n(K(c)) \rightarrow H_n(K(d))$$

for every  $n$  if every  $\Theta_{\alpha,n}$  vanishes.

**Proof.** If  $z \in \ker \partial_n^c$ , then

$$\partial_n^d T_{\alpha,n} z = T_{\alpha,n-1} \partial_n^c z = 0,$$

so cycles map to cycles. If  $z = \partial_{n+1}^c w$ , then

$$T_{\alpha,n} z = T_{\alpha,n} \partial_{n+1}^c w = \partial_{n+1}^d T_{\alpha,n+1} w,$$

so boundaries map to boundaries. The quotient map is therefore well-defined.  $\square$

### Corollary 2.3 — Exact failure witness

When the complexes are finite and matrices are represented exactly,  $\Theta_{\alpha,n} \neq 0$  has a finite basis-vector witness. Choose a nonzero column  $j$ ; then

$$\Theta_{\alpha,n} e_j \neq 0.$$

The index  $j$  identifies a source obligation whose verification residual changes under transport. The residual vector is an independently checkable counterexample to naturality.

### Theorem 2.4 — Defect derivation law

For composable transports  $T_\alpha : c \rightarrow d$  and  $T_\beta : d \rightarrow e$ ,

$$\Theta_{\beta\alpha} = \Theta_\beta T_\alpha + T_\beta \Theta_\alpha.$$

**Proof.** Expand

$$\partial^e T_\beta T_\alpha - T_\beta T_\alpha \partial^c$$

and add and subtract  $T_\beta \partial^d T_\alpha$ .  $\square$

The law localizes composite failure. Cancellation between stage defects is possible, but the cancellation itself becomes a checkable identity and cannot be inferred from a favorable endpoint alone.

## 2.4 Path holonomy

Suppose two declared paths from  $c$  to  $d$  are

$$c \xrightarrow{p} c_p \xrightarrow{q|p} d, c \xrightarrow{q} c_q \xrightarrow{p|q} d.$$

Define

$$\Omega_{p,q} = T_{q|p} T_p - T_{p|q} T_q.$$

Exact path independence is  $\Omega_{p,q} = 0$ . A nonzero column is a path-dependent certificate witness. In the arithmetic program,  $p$  and  $q$  can denote the order of adjoining primes. In the trans-scale program, they can denote two scale refinements. In an observation pipeline, they can denote quotienting and changing instruments.

Triangle tests alone do not detect every such defect. A context graph may contain an unfilled square and no triangles, so every triangle condition is vacuous while  $\Omega_{p,q} \neq 0$ . Coherence must be checked on a declared generating family of two-cells or path rewrites.

## 2.5 Curved audit bicomplex

Let  $K_{p,q}$  carry a horizontal differential  $\delta: K_{p,q} \rightarrow K_{p-1,q}$  and a vertical differential  $\partial: K_{p,q} \rightarrow K_{p,q-1}$ , with  $\delta^2 = \partial^2 = 0$ . On  $K_{p,q}$ , set

$$D = \delta + (-1)^p \partial.$$

Then

$$D^2 = (-1)^p (\delta \partial - \partial \delta).$$

Thus the audit curvature is the interchange commutator. Only a flat double complex has a genuine total homology without correction. This supplies an exact replacement for distance-valued “cohomology” language: a cohomology theory needs typed coefficients and a coboundary; a path distance by itself is not a cocycle.



### 3. Observation-descent Galois lattice

#### 3.1 Exact observation relations

Let  $X$  be a raw state set. An exact observation relation  $R \subseteq X \times X$  identifies states that an instrument or legal filtration cannot distinguish. For a query  $q: X \rightarrow Y_q$ , define

$$\ker q = \{(x, x') \in X^2 : q(x) = q(x')\}.$$

For a query family  $Q$ , define

$$\text{Ker}(Q) = \bigcap_{q \in Q} \ker q.$$

For a relation  $R$ , define its descending queries

$$\text{Desc}(R) = \{q : R \subseteq \ker q\}.$$

#### Theorem 3.1 — Observation/query Galois connection

$$Q \subseteq \text{Desc}(R) \Leftrightarrow R \subseteq \text{Ker}(Q).$$

**Proof.** Both statements assert that every  $q \in Q$  is constant on every pair in  $R$ . ▢

The maps  $\text{Ker}$  and  $\text{Desc}$  are antitone. Their composites are closure operators on declared relations and query families. A failed descent has a concrete witness

$$(x, x') \in R, q(x) \neq q(x').$$

This is the executable form of quotient legality.

#### Corollary 3.2 — Informativeness order

If  $R_1 \subseteq R_2$ , then

$$\text{Desc}(R_2) \subseteq \text{Desc}(R_1).$$

An experiment that identifies fewer state pairs admits at least as many well-defined queries.

#### 3.2 Quotient caution

Even when  $X$  is a standard Borel space and  $R$  is a Borel equivalence relation, the quotient need not inherit a standard Borel structure. The court should emit a quotient object only when a smooth classification, countable separating family, or explicit code is supplied. Otherwise the correct object is the relation, its kernel-pair groupoid, or an identified set.

Approximate confusability is generally not transitive. A tolerance relation  $d(x, x') \leq \varepsilon$  can have  $x$  close to  $y$  and  $y$  close to  $z$  while  $x$  is not close to  $z$ . Such data should remain a pseudometric ball system, weighted graph, or uncertainty set; forcing it into a quotient silently changes the mathematics.

Observational equivalence is also weaker than interventional equivalence. A causal claim requires a declared intervention family and equality or controlled discrepancy of interventional laws, not merely equal passive observations.

## 4. Quantitative defect transport

The exact chain theory detects whether a certificate commutes. Approximate scientific systems additionally require controlled defect propagation.

Let  $(X, d_X)$  and  $(Y, d_Y)$  be Polish metric spaces, and let  $K : X \rightsquigarrow Y$  be a Markov kernel whose output laws have finite first moment. Define the Wasserstein-Dobrushin modulus

$$L(K) = \sup_{x \neq x'} \frac{W_1(K_x, K_{x'})}{d_X(x, x')}.$$

For parallel kernels  $K, L : X \rightsquigarrow Y$ , define

$$\Delta(K, L) = \sup_{x \in X} W_1(K_x, L_x).$$

### 4.1 Two-stage defect transport

For  $K, L : X \rightsquigarrow Y$  and  $M, N : Y \rightsquigarrow Z$ ,

$$\Delta(M \circ K, N \circ L) \leq L(M) \Delta(K, L) + \Delta(M, N).$$

**Proof.** Insert  $M \circ L$  and apply the triangle inequality. The map sending an input measure  $\mu$  on  $Y$  to  $\mu M$  is  $L(M)$ -Lipschitz in  $W_1$ , giving the first term. Integrating the pointwise bound  $W_1(M_y, N_y) \leq \Delta(M, N)$  against  $L_x$  gives the second. Taking the supremum over  $x$  proves the claim.  $\square$

### 4.2 Affine certificate monoid

Encode a certified stage by a triple  $(L, \varepsilon, \alpha)$ , where  $L \geq 0$  bounds amplification,  $\varepsilon \geq 0$  bounds newly introduced defect, and  $0 \leq \alpha \leq 1$  bounds certificate failure probability. Define

$$(L_2, \varepsilon_2, \alpha_2) \star (L_1, \varepsilon_1, \alpha_1) = (L_2 L_1, \varepsilon_2 + L_2 \varepsilon_1, \min\{1, \alpha_1 + \alpha_2\}).$$

#### Proposition 4.2

The operation  $\star$  is associative with identity  $(1, 0, 0)$ .

**Proof.** Multiplication is associative in the first coordinate. Both bracketings give

$$\varepsilon_3 + L_3 \varepsilon_2 + L_3 L_2 \varepsilon_1$$

in the second coordinate. Truncated addition on  $[0, 1)$  is associative in the third.  $\square$

This monoid is deliberately conservative. The probability coordinate uses the union bound and assumes no independence. The audit engine computes the composite over exact rationals and rejects any declared path certificate that understates a coordinate.

## 5. Arithmetic atomic rigidity

### 5.1 The finite matrix boundary

For a finite self-adjoint matrix  $A$ , the distributional trace of its unitary flow paired against a test function has the form

$$f \mapsto \sum_{j=1}^N c_j \hat{f}(\lambda_j),$$

equivalently a distribution with a finite analytic exponential density in the flow variable. It cannot equal the infinite locally finite prime-power comb on  $C_c^\infty(\mathbb{R})$ . A counterterm allowed to contain that comb would make the operator contribution arithmetically nonidentifiable. Therefore an exact prime law requires an infinite-dimensional distributional trace or a statement explicitly restricted to a finite truncation.

This no-go theorem is a permanent gate, not a disappointment. It eliminates zero-fitting and naive finite matrix reconstruction before spectral computation begins.

### 5.2 Admissible counterterm class

Let  $X$  consist of distributions of the form

$$T = C_0 + a(u) du + \sum_{t \in E} c_t \delta_t,$$

where  $\text{supp } C_0 \subseteq \{0\}$ ,  $a \in L_{loc}^1(\mathbb{R})$ , and  $E \subset \mathbb{R} \setminus \{0\}$  is locally finite. The off-origin atomic decomposition is unique. Define the atomic charge

$$At(T) = \sum_{t \in E} c_t [t]$$

in the free complex vector space on  $\mathbb{R} \setminus \{0\}$ .

Admissible counterterms have the form  $C_0 + a(u) du$ , so their atomic charge vanishes. A finite analytic exponential trace also has vanishing atomic charge.

For a rational prime  $p$ , define the symmetric prime-power comb

$$P_p = \sum_{m \geq 1} \frac{\log p}{p^{m/2}} (\delta_{m \log p} + \delta_{-m \log p}).$$

Then

$$At(P_p) = \sum_{m \geq 1} \frac{\log p}{p^{m/2}} ([m \log p] + [-m \log p]).$$

### Proposition 5.1 — Prime atomic independence

The family  $\{At(P_p); p \text{ prime}\}$  is linearly independent.

**Proof.** The supports for distinct primes are disjoint. If  $m \log p = n \log q$  for rational primes  $p, q$ , then  $p^m = q^n$ . Unique factorization gives  $p = q$ , and then  $m = n$ . The same holds for negative locations. A finite linear combination can vanish only if every prime coefficient vanishes.  $\square$

Atomic charge therefore records exactly which prime channels occur. It is invariant under every declared admissible counterterm and is stronger than a finite test-function battery.

## 5.3 The concentration loophole

Stagewise admissibility is not stable under limits. Let  $\eta \in C_c^\infty(-1, 1)$  be nonnegative with  $\int \eta = 1$ . For a fixed prime  $p$ , set

$$a_R(u) = \sum_{m \leq R} \frac{\log p}{p^{m/2}} R [\eta(R(u - m \log p)) + \eta(R(u + m \log p))].$$

For every finite  $R$ ,  $a_R(u) du$  is a smooth, absolutely continuous measure and has zero atomic charge. Yet

$$a_R(u) du \rightarrow P_p$$

distributionally. A counterterm channel constrained only stage by stage can therefore reconstruct the forbidden comb in the joint cutoff limit.

## 5.4 Concentration defect

For a compact  $K$  disjoint from the origin and a directed family  $a_\lambda(u) du$ , define

$$\chi_K = \lim_{\varepsilon \downarrow 0} \lim_{\lambda} \sup_{t \in K} \sup_{t-\varepsilon}^{t+\varepsilon} \int |a_\lambda(u)| du.$$

The operational certificate is a computable modulus  $\kappa_K(\varepsilon)$  such that, eventually in  $\lambda$ ,

$$\sup_{t \in K} \int_{t-\varepsilon}^{t+\varepsilon} |a_\lambda(u)| du \leq \kappa_K(\varepsilon), \kappa_K(\varepsilon) \rightarrow 0.$$

A power certificate  $\kappa_K(\varepsilon) \leq C_K \varepsilon^{\gamma_K}$ , with  $\gamma_K > 0$ , is directly checkable once the uniform analytic bound is supplied.

### Theorem 5.2 — Atomic rigidity

Let  $\mu_\lambda = a_\lambda(u) du$  converge weakly as complex Radon measures on compact sets disjoint from the origin to  $\mu$ . Assume local uniform boundedness of total variation and the vanishing concentration modulus above. Then  $\mu$  has no atom away from the origin.

**Proof.** Fix  $t \neq 0$  and a compact neighborhood  $K$  of  $t$  disjoint from the origin. For sufficiently small  $\varepsilon$ , choose continuous functions  $\varphi_r$ ,  $0 \leq \varphi_r \leq 1$ , equal to one on  $[t-r, t+r]$ , supported in  $[t-\varepsilon, t+\varepsilon]$ , and decreasing pointwise to the indicator of  $\{t\}$  as  $r \downarrow 0$ . Weak convergence gives

$$\left| \int \varphi_r d\mu \right| = \lim_{\lambda} \left| \int \varphi_r d\mu_\lambda \right| \leq \lim_{\lambda} \sup_{t \in K} \int_{t-\varepsilon}^{t+\varepsilon} |a_\lambda(u)| du \leq \kappa_K(\varepsilon).$$

Local finite variation and dominated convergence yield  $\int \varphi_r d\mu \rightarrow \mu(\{t\})$ . Hence  $|\mu(\{t\})| \leq \kappa_K(\varepsilon)$ . Let  $\varepsilon \downarrow 0$ .  $\square$

### Corollary 5.3 — Raw-origin theorem

If a limiting trace identity contains  $P_p$ , while its counterterm family satisfies atomic rigidity away from zero, then the prime atomic charge must originate in the raw trace channel.

This is the bridge between the finite prime ledger and the infinite-dimensional operator fold: both must expose the same canonical atomic signature.

## 5.5 A sufficient regularity certificate

On a compact interval  $I$ , a uniform  $W^{1,1}(I)$  bound supplies a linear concentration modulus. For an absolutely continuous representative,

$$\|a\|_\infty \leq |I|^{-1} \|a\|_1 + \|a'\|_1.$$

Therefore

$$\sup_t \int_{t-\varepsilon}^{t+\varepsilon} |a(u)| du \leq 2\varepsilon (|I|^{-1} \|a\|_1 + \|a'\|_1).$$

The checker uses the compact exhaustion

$$K_n = [-n, -1/n) \cup [1/n, n).$$

Passing finite sample records establishes only their internal consistency. The uniform analytic proof remains a separate certificate obligation.

## 6. Consequences for the arithmetic program

### 6.1 Phase 1 — Effective rational dictionary

Work in the logarithmic half-density variable  $u = \log x$ . The dictionary should enumerate rational compactly supported splines and their convolutions with rational translates and dilates of one fixed normalized mollifier in  $C_c^\infty(\mathbb{R})$ .

Two certificates must be separated.

**DictionaryAtomCertificate.** Records the enumeration index, spline degree, rational knots and coefficients, mollifier identifier and scale, inset support, reflection partner, and explicit derivative bounds.

**ApproximationCertificate.** Begins only after a target  $f$ , support stage  $m$ , derivative order  $k$ , and requested error  $\eta$  are declared. It records a finite atom combination and separate boundary, spline, rationalization, and mollification errors.

An executable density theorem requires one explicit quasi-interpolant and computable constants. A notebook can verify instances; it does not prove LF density merely by plotting convergence. Fair enumeration should order parameter tuples by bounded total height so that support stage, degree, mesh, mollification scale, derivative order, and coefficient height all advance.

### 6.2 Phase 2 — Interval Weil Gram engine

For logarithmic supports  $[a_j, b_j]$  and  $[a_i, b_i]$ , the convolution with reflected conjugate has certified outer support

$$\text{supp}(\phi_j * \phi_i^*) \subseteq [a_j - b_i, b_j - a_i].$$

This is an **outer support**, sufficient for the arithmetic cutoff. It is not automatically sharp; sharpness requires endpoint-cancellation analysis or a Titchmarsh-type certificate.

For exact rational outer radius  $L$ , an enclosure of  $e^L$  must be refined until its floor is isolated. Prime powers are then enumerated by a complete integer sieve and exact integer multiplication, with primality and completeness transcripts. Arithmetic, pole, removable-origin, compact archimedean, and analytic-tail contributions are enclosed separately. Entries  $(i, j)$  and  $(j, i)$  are computed independently and intersected after conjugation. Empty intersection is a contradiction, never an invitation to average.

At singular Gram stages, the preferred certificate is an exact function relation

$$\sum_j Z_{j\alpha} \phi_j = 0.$$

Sesquilinearity then proves  $G_N Z = 0$  and  $Z^* b_N = 0$ . A rational complement  $U$ , invertibility of  $[U Z]$ , and positivity of  $U^* G_N U$  replace a floating pseudoinverse. Zero tables remain outside the primary arithmetic calculation and belong only to a separate spectral audit.

### 6.3 Phase 3 — Infinite-dimensional fold

An admissible operator proposal must provide:

1. a separable Hilbert space and explicitly typed local prime objects;
2. self-adjointness or a certified self-adjoint realization;
3. intertwiners between finite-place stages;
4. relative resolvent differences in the trace class;
5. exact distributional prime increments, not only finite-battery agreement;
6. origin-jet or controlled absolutely continuous counterterms;
7. prime-order holonomy control;
8. atomic-rigidity moduli for counterterm nets;
9. a joint Cauchy modulus for place and cutoff limits;
10. a determinant normalization compatible with the limiting trace identity.

Passing a finite battery can refute a distributional identity but cannot prove equality on  $C_c^\infty(\mathbb{R})$ . It receives the state **strongly supported**, never **proven**.

Recent spectral constructions based on scaling operators, rank-one perturbations, and truncated Weil forms are important neighboring programs. Their reported numerical precision does not settle the convergence required here; the relevant preprints explicitly leave that convergence open. The audit court therefore treats them as hypothesis-generating benchmarks, not operator certificates.

## 7. Catalogue-wide mathematical corrections

The following repairs preserve the strongest parts of the framework while removing category leakage.

### 7.1 Cohomology requires coefficients

A distance-valued triangle defect is not by itself cohomology. A legitimate cohomology claim must specify a coefficient group or sheaf, cochains, a coboundary, and the equivalence relation on cocycles. If

the data are genuinely nonabelian, an alleged second Čech cohomology class needs a typed crossed module, central extension, gerbe, or higher-categorical replacement. Otherwise the honest object is a defect two-category with explicit path witnesses.

## 7.2 Trace bounds require the correct direction

Operator norm bounds provide upper estimates:

$$\|Tx\| \leq \|T\| \|x\|.$$

A lower bound requires injectivity and a positive minimum singular value on a declared subspace. Any “trace extension lower bound” inferred solely from an operator norm is invalid.

## 7.3 Local nondegeneracy is not global recovery

A nonzero Jacobian determinant proves local invertibility under the inverse function theorem. It does not prove global injectivity, properness, or boundary completeness. A global-recovery certificate must separately control fibers and escape to infinity. The audit engine implements this as a dedicated gate.

## 7.4 Boundary sufficiency is vector-valued

A single scalar can hide a large exterior residual. Retain at least the ordered pair

$$\left( I(Y; X_\partial \mid B), I(Y; X_{ext} \mid X_\partial, B) \right).$$

The first coordinate measures information captured at the boundary; the second measures information still present outside it after conditioning. Compression to one score requires a preregistered decision rule.

## 7.5 Physical conservation ledgers

Poynting, energy, and flux checks should return an interval enclosure of the complete residual after calibration, windowing, detector response, unmeasured channels, and numerical error. Inconsistent component intervals must not be averaged. Optical analogue-horizon experiments demonstrate measurable frequency conversion and backreaction architecture; they do not imply a prime-local arithmetic operator or a distributional trace theorem.

## 7.6 Causal and simulation claims

Intensity detection can quotient global phase, while nonlinear dynamics can remain phase-sensitive. That distinction is physically useful. It does not identify a hidden computational grid. A simulation-resolution claim needs a registered null family containing dispersion, Kerr corrections, Raman effects, self-steepening, loss, detector leakage, and numerical discretization, plus an identifiable parameter



whose confidence set excludes the null. Without this, “scale torsion” is an analogy, not a measured invariant.

## 7.7 Scores require units

Weighted mechanism, rent, complexity, or human-impact scores are not invariant under rescaling their components. Either preregister nondimensionalization and weights before evaluation or retain the Pareto vector. Normative and civic volumes can guide deployment policy, but they are not promoted as mathematical theorems without additional formalization.

# 8. Executable audit engine

## 8.1 Architecture

The repository implements a small trusted kernel rather than a monolithic scientific oracle.

| Layer              | Current function                                          | Certificate boundary             |
|--------------------|-----------------------------------------------------------|----------------------------------|
| Manifest court     | Types claims, statuses, gates, demotion, and preservation | Parsing does not infer truth     |
| Exact algebra      | Rational matrices, rank, complexes, transports            | Floating scalars are rejected    |
| Descent court      | Interchange defects and path holonomy                     | Emits basis-vector residuals     |
| Observation court  | Relation/query descent                                    | Emits distinguishing state pairs |
| Product gate court | Conflict preservation and fatal propagation               | No aggregate rescue              |
| Defect court       | Exact affine bound composition                            | Rejects understated path budgets |
| Atomic court       | Finite nonconcentration records                           | Uniform proof remains external   |

| Layer             | Current function                                          | Certificate boundary                     |
|-------------------|-----------------------------------------------------------|------------------------------------------|
| Arithmetic plugin | Finite-comb, zero-fit, counterterm, and trace-limit gates | Does not build the operator              |
| Recovery plugin   | Local/global promotion audit                              | Requires fiber and boundary certificates |
| Provenance        | Canonical JSON and SHA-256 identifiers                    | Hashes identify bytes, not validity      |

The producer should eventually be allowed to use heavy interval libraries. The independent verifier should remain small, offline, deterministic, and capable of checking emitted rational witnesses, interval inclusions, sieve completeness, hashes, and verdict logic.

## 8.2 Certificate envelope

Every artifact should be wrapped in a versioned envelope:

```

CertificateEnvelopeV1
  schema_version
  kind
  claim_id
  formula_spec_hash
  assumptions[]
  inputs[]
  payload
  producer_build
  dependency_lock_hash
  rounding_policy
  created_utc

```

The producer does not assign its own theorem verdict. The current verifier emits a mechanical audit decision, structured findings, and an actual-execution ledger. A theorem is mechanically admissible only when a claim-bound, hash-verified proof artifact reports a passing result and every hard gate is derived as passing from the complete bound evidence set. Human research classifications such as proven, refuted, supported, or unresolved remain separate from the engine's finite checks unless a declared verification adapter supplies the corresponding certificate.

### 8.3 Implemented commands

```
bsc-audit lint    claim.json
bsc-audit audit   claim.json
bsc-audit complex transport.json
bsc-audit observe observation.json
bsc-audit atomic  concentration.json
bsc-audit defect  path_bounds.json
```

Every command emits canonical JSON findings and an actual-execution ledger. Exit code zero means that no blocking finding was produced; exit code one means that a valid input was blocked or demoted; exit code two means malformed input or command usage; and exit code seventy reports an internal verifier failure. Checks skipped after a fail-closed structural error are recorded as not run, never inferred as executed.

### 8.4 Adversarial benchmark suite

The next build should preserve at least these negative fixtures:

- omit one prime power from an otherwise correct ledger;
- mark a composite as prime or omit sieve-completeness evidence;
- use outer support as if sharp after endpoint cancellation;
- average nonoverlapping Hermitian boxes;
- project an indefinite matrix onto the positive cone;
- cross a singular stage with a floating pseudoinverse;
- import a zero table into a primary Gram dependency graph;
- fit a counterterm after inspecting spectral output;
- hide a delta comb through narrowing smooth bumps;
- substitute strong-resolvent convergence for a trace-class estimate;
- pass a finite function battery and request a distributional verdict;
- produce order-matched spectra without an intertwiner;
- let the two cutoff orders yield different determinant limits;
- declare an approximate confusability graph to be an equivalence relation;
- admit a root claim while one fatal coordinate is unrun, failed, or conflicting;
- understate an affine path-defect budget.

## 9. Novelty audit

The mathematical ingredients have established ancestors:

- chain complexes, chain maps, homology, and double complexes are standard;
- Galois connections between relations and invariant functions are standard order theory;
- Markov categories, comparison of statistical experiments, Wasserstein metrics, and Dobrushin coefficients are established;
- uniform integrability and nonconcentration criteria are classical measure theory;
- distributional trace formulae and spectral approaches to zeta zeros have a substantial literature.

The candidate contribution is their **typed assembly into an executable audit-descent calculus** with four linked invariants:

1. certificate-interchange defect  $\Theta$ ;
2. context path holonomy  $\Omega$ ;
3. observation descent closure  $(Ker, Desc)$ ;
4. off-origin atomic charge plus concentration defect  $\chi_K$ .

The arithmetic application adds a specific new design boundary: stagewise admissible counterterms must carry a uniform atomic-rigidity certificate before a joint limit is accepted. The same prime atomic ledger then constrains both finite arithmetic evaluation and infinite-dimensional trace ablation.

Historical priority for this exact assembly has not been established. The correct claim state is therefore **candidate theorem package with proved internal identities and unresolved novelty priority**.

## 10. Failure modes and falsifiers

The proposal should be rejected or narrowed if any of the following occurs.

1. A claimed context transport lacks typed maps at every certificate degree.
2. A nonzero  $\Theta$  is hidden by checking only the final numerical output.
3. Path independence is inferred from triangles when the context complex contains unfilled squares.
4. Approximate transport is reported without a metric, norm, domain, and tolerance.
5. A quotient query lacks a descent proof or distinguishing-pair search.
6. A Borel quotient is asserted standard without a smoothness or coding certificate.
7. A fatal gate is treated as a scalar penalty or an operator ideal.
8. A counterterm family is checked only stagewise, with no uniform nonconcentration modulus.

9. Finite samples are presented as proof of a uniform analytic bound.
10. A finite matrix or zero-fitted boundary condition is claimed to realize the full prime comb.
11. A spectral fit is promoted despite absent intertwiners, trace-class bounds, or joint convergence.
12. Independent implementations disagree on an exact witness, hash, or enclosure.

## 11. Publication readiness

The exact algebraic core is ready for a short foundations preprint after independent proof review and a repository release. The atomic-rigidity section is ready as a theorem note, provided the declared topology of convergence and measure class are retained exactly. The catalogue-wide synthesis is suitable as a research-program paper, not as a completed theory of physics.

The arithmetic dictionary and Gram engine should be published only after certificate files can be regenerated from a clean environment and checked by a second implementation. The infinite-dimensional operator fold remains a frontier problem. No claim about the Riemann hypothesis should appear in a title, abstract, or conclusion unless the missing self-adjointness, exact increment, and joint trace-class convergence obligations are actually discharged.

## 12. Highest-leverage next step

Implement one end-to-end **Prime Atom Preservation Packet** before enlarging the theory.

For a small declared support radius and two primes, the packet should contain:

1. exact dictionary atom records and inset-support certificates;
2. an exact convolution outer-support record;
3. a complete prime-power ledger with independent sieve transcript;
4. one rigorously enclosed Weil Gram entry and its independently computed transpose entry;
5. an off-origin atomic-charge record for the corresponding raw prime increment;
6. a counterterm family certificate on  $K_1, K_2, K_3$  with explicit  $W^{1,1}$  bounds;
7. both prime-adjunction orders and an exact or bounded holonomy certificate;
8. a content-addressed preservation manifest;
9. positive and intentionally corrupted fixtures;
10. verification by the small offline checker.

This packet joins the finite arithmetic ledger to the infinite trace boundary without pretending that the operator already exists. If it succeeds, it becomes the minimal executable theorem object from which Phase 1, Phase 2, and Phase 3 can grow coherently.

## 13. Open problems

1. Construct a canonical certificate complex for LF test-function approximation whose homology has a useful analytic interpretation.
2. Determine minimal generating two-cells for the scale/instrument/quotient context category.
3. Develop interval-valued chain complexes in which enclosures compose without losing decisive witnesses.
4. Characterize when observation-descent closure preserves standard Borel realizability.
5. Replace union-bound probability composition by sharper certificates when dependence information is itself certified.
6. Find the weakest effective condition that excludes off-origin atomic concentration for distribution-valued counterterms beyond measures.
7. Determine whether atomic charge extends naturally to a quotient of tempered distributions suitable for the full explicit formula.
8. Construct prime-local Hilbert stages whose raw trace increments carry the prescribed atomic charge before counterterms.
9. Prove a joint place/cutoff interchange theorem in trace norm, or compute a rigorous naturality-defect modulus.
10. Formalize the finite-comb no-go theorem, support calculus, Schur gate, and atomic-rigidity theorem in a proof assistant.
11. Establish historical priority and precise relation to categorical proof theory, institution theory, and formalized scientific workflows.
12. Test whether the audit-descent calculus compresses real multi-instrument scientific disagreements better than ordinary provenance graphs.

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## Appendix A. Minimal manifest fragment

```
{
  "manifest_version": "0.3.0",
  "draft": true,
  "claim": {
    "id": "example:claim",
    "title": "Example research claim",
    "type": "conjecture",
    "evidence_maturity": "declared",
    "deployment_status": "research_only",
    "statement": "A precisely scoped example claim remains unresolved.",
    "scope": "illustrative manifest only"
  },
  "system": {"domain": "declared system", "state_type": "typed states"},
  "observation": {"kernel_or_instrument": "declared instrument", "legal_filtration": {}},
  "representation": {"kind": "identity"},
  "target": {"outcome": "declared outcome", "horizon": "declared horizon", "loss_or_score": "declared score"},
  "experiment": {"baseline_model": "ordinary baseline", "search_budget": "not started"},
  "admission": {
    "hard_gates": ["naturalness"],
    "gate_results": [{"id": "naturalness", "state": "unrun", "fatal": true, "evidence": []}]
  },
  "demotion": {
    "owner": "maintainer",
    "rules": [{"if": "counterexample", "then": "retire"}],
    "negative_result_destination": "negative-results/"
  },
  "preservation": {"known_failures": ["No proof or experiment has been supplied."]},
  "evidence": []
}
```

## Appendix B. Engine verification record

The version 0.3.0a2 engine is tested under pinned Python 3.12.13 and setuptools 82.0.1. Its 62 deterministic unit tests cover exact arithmetic, schema/runtime parity, evidence-derived gate states, conflict and omission detection, failed-proof rejection, fail-closed arithmetic configuration, fatal propagation, and certificate calculations. A dedicated Null-Discrimination suite preserves the four known false-pass cases as regression fixtures.

Continuous integration is a build fact, not independent replication; the latter requires an independently operated verifier and claim-bound passing evidence.